

PARMA, Italy

WMO: 161300

Lat: 44.8172N

Lon: 10.2969E

Elev: 51

StdP: 100.71

Time Zone: 1.00 (EUC)

Period: 06-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	-3.9	-9.3	1.7	-1.8	-7.4	2.0	-0.5	8.2	10.5	5.8	8.6	1.5	200	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	34.0	21.6	32.8	21.5	31.1	21.0	23.1	30.7	22.5	30.2	21.8	29.4	2.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	15.6	27.0	20.0	14.8	26.1	19.1	14.0	25.4	68.7	30.9	66.4	30.2	64.0	29.3	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.5	6.1	5.0	-8.5	36.5	3.4	1.2	-11.0	37.3	-13.0	38.0	-14.9	38.7	-17.4	39.5		
			-8.7	24.5	3.6	0.6	-11.3	24.9	-13.4	25.3	-15.4	25.7	-18.0	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.1	3.2	4.7	9.2	13.9	18.2	22.2	24.8	24.1	20.0	15.0	9.4	3.6
	DBStd	8.33	3.86	4.33	3.48	2.95	3.00	3.12	2.40	2.71	2.52	3.83	3.90	3.86
	HDD10.0	688	215	154	56	4	0	0	0	0	0	6	52	201
	HDD18.3	2161	469	382	282	134	41	5	0	1	11	113	268	457
	CDD10.0	2184	5	6	32	122	254	367	459	438	300	162	35	3
	CDD18.3	614	1	1	1	2	36	122	201	179	61	11	1	0
	CDH23.3	5956	0	0	5	33	324	1147	2147	1831	433	35	0	0
	CDH26.7	2305	0	0	0	3	73	419	929	779	98	3	0	0
Wind	WSAvg	2.0	1.5	1.8	2.4	2.3	2.5	2.3	2.3	2.2	1.9	1.7	1.6	1.6
Precipitation	PrecAvg	923	63	67	76	81	70	65	49	81	78	103	116	68
	PrecMax	1188	150	163	277	220	151	180	129	187	187	326	211	131
	PrecMin	665	2	1	3	12	12	8	5	0	2	0	15	4
	PrecStd	154	46	41	61	49	42	45	36	58	51	89	51	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.7	19.7	22.9	26.2	31.2	33.9	35.5	36.1	31.0	26.8	19.7	19.7
		MCWB	9.4	12.8	12.6	16.4	20.4	21.9	21.3	21.2	21.0	17.6	16.5	18.0
	2%	DB	19.3	19.3	19.9	24.0	29.0	32.6	34.1	34.1	29.2	23.9	19.7	19.5
		MCWB	10.0	12.4	12.1	15.6	18.4	21.5	21.6	21.5	20.1	17.9	16.7	18.0
	5%	DB	9.8	12.7	17.9	22.2	27.0	31.0	32.8	32.4	27.6	21.9	19.5	11.9
		MCWB	7.2	8.9	11.4	14.7	17.9	21.3	21.6	21.2	19.0	17.1	17.0	9.0
	10%	DB	7.6	10.2	15.9	20.4	25.0	29.1	31.1	31.0	25.9	20.1	19.3	8.9
		MCWB	6.6	7.2	10.7	14.2	16.8	20.4	21.2	20.6	18.0	15.9	17.4	7.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	10.9	13.9	17.6	21.0	23.9	24.0	23.8	22.1	19.6	18.2	18.2
		MCDB	14.9	13.9	19.4	23.9	28.4	30.6	31.5	31.3	28.1	22.8	19.2	19.4
	2%	WB	8.4	9.7	13.0	16.4	19.6	22.7	23.2	23.0	21.0	18.6	18.2	18.0
		MCDB	9.8	12.2	18.3	21.7	26.5	30.3	30.9	30.6	26.9	21.5	19.2	19.0
	5%	WB	7.3	8.6	12.2	15.5	18.7	21.8	22.5	22.3	20.2	18.2	18.0	9.9
		MCDB	8.2	10.6	17.0	20.9	25.2	29.4	30.4	29.9	25.8	20.2	19.0	11.1
	10%	WB	6.3	7.8	11.2	14.7	17.6	20.9	21.8	21.5	19.4	17.9	17.8	7.6
		MCDB	7.0	9.3	14.8	19.9	23.7	28.1	29.6	29.0	24.3	19.7	18.8	8.5
Mean Daily Temperature Range	5% DB	MDBR	5.9	8.0	9.9	10.3	11.2	10.9	11.9	11.7	9.5	7.7	6.7	7.5
		MCDBR	10.4	10.6	13.8	13.9	13.4	12.9	13.4	13.7	12.5	10.3	9.5	13.8
		MCWBR	7.2	8.0	7.9	7.4	6.1	5.6	5.2	5.5	5.5	5.5	7.2	11.5
	5% WB	MCDBR	8.4	9.7	11.9	12.0	12.2	12.3	12.2	12.1	10.2	6.7	10.0	13.0
		MCWBR	6.2	7.1	7.2	6.7	6.1	5.5	5.1	5.3	4.8	4.1	9.7	12.0
Clear-Sky Solar Irradiance	taub	0.361	0.390	0.437	0.452	0.448	0.469	0.454	0.444	0.433	0.438	0.390	0.355	
	taud	2.334	2.220	2.094	2.099	2.156	2.145	2.194	2.236	2.238	2.169	2.255	2.367	
	Ebn at Noon	731	773	783	809	826	808	815	809	782	706	680	698	
	Edh at Noon	81	108	140	151	147	149	141	130	120	113	87	72	
All-Sky Solar Radiation	RadAvg	1.45	2.25	3.52	4.53	5.63	6.46	6.66	5.70	4.21	2.53	1.50	1.24	
	RadStd	0.21	0.34	0.51	0.51	0.53	0.37	0.37	0.38	0.28	0.36	0.22	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	+0.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+64	+14

Nomenclature: See separate page